

SMS Shield: Spam Detection System for Short SMS Messages

SMS Shield is a machine learning web application designed to classify short text messages as either spam or legitimate messages, referred to as ham. The project focuses on the practical challenges of detecting spam in short, noisy SMS data where traditional word based methods often struggle. The system provides a clean, responsive interface where a user can paste a message, receive an immediate classification, and review a confidence score generated by the model. If the prediction is incorrect, the user can submit the correct label, which is then stored for future retraining. This creates a feedback driven improvement loop that reflects how real world classification systems evolve over time.

The core objective of this project was to build an end to end machine learning pipeline that integrates model training, evaluation, deployment, and user feedback into a single cohesive system. The training data comes from the UCI SMS Spam Collection dataset, which contains 5,574 labeled English SMS messages categorized as either ham or spam. Spam examples in this dataset include promotional offers, phishing attempts, suspicious links, prize notifications, and other deceptive content. Ham messages include normal conversations, reminders, and personal communication.

Several classification approaches were evaluated during development. Word level TF IDF with Multinomial Naive Bayes was implemented as a baseline model. A linear Support Vector Machine was also tested to compare performance. While both models performed reasonably well on clear promotional spam, they showed limitations when handling obfuscated text and short messages with altered spelling. The final model selection was based on character n gram TF IDF features combined with Logistic Regression. Character level features capture patterns within

words, allowing the model to detect variations such as fr33 instead of free, cl1ck instead of click, and other common spam obfuscation techniques. Logistic Regression provided stable probability outputs, which enabled the system to display meaningful confidence scores to users.

The application is built using Python and Flask for the backend, with scikit learn handling feature extraction and classification. The trained model is stored using joblib and loaded at runtime to provide fast predictions. The frontend is developed with HTML, CSS, and vanilla JavaScript, featuring a modern dark themed interface, responsive layout, character counter, status indicator, and real time feedback elements. The system is structured to maintain separation between templates, static assets, model files, and training logs, following standard web application design principles.

A key feature of SMS Shield is its feedback mechanism. After receiving a prediction, users can confirm or correct the label. The confirmed message and label are stored in a CSV file, creating a growing dataset of verified examples. This supports future retraining and continuous improvement of the classifier. The confidence score presented in the interface represents the probability output from the Logistic Regression model and serves as a signal of prediction strength rather than absolute certainty.

SMS Shield demonstrates practical skills in machine learning model evaluation, feature engineering, deployment with Flask, user interface design, and feedback driven data collection. The project reflects an applied understanding of text classification challenges in real world environments and provides a foundation for future enhancements such as evaluation dashboards, adjustable decision thresholds, model explainability features, and automated retraining workflows.